

WolfLLM: Training and Evaluating Large Language Models for Werewolf Social Deduction

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Abstract

Large Language Models (LLMs) have achieved strong performance on static reasoning benchmarks, but their ability to reason and communicate in dynamic social environments with incomplete information remains underexplored. In this paper, we study LLMs in Werewolf, a social deduction game that requires long-context tracking, identity inference, player-perspective reasoning, and strategic communication. We construct a real-game Werewolf corpus by converting multi-speaker game videos into structured trajectories with player numbers, stages, roles, public events, and key actions. To preserve the information structure of the game, we reconstruct player-perspective information states so that each sample only contains information available to the current player. Based on this data, we build a dedicated benchmark with two tasks: identity inference and speech generation, and train WolfLLM through supervised fine-tuning. Experiments show that WolfLLM substantially improves identity inference over the Qwen3-0.6B base model, increasing overall field-level accuracy from 1.08% to 56.18%, and achieves the best overall performance among the evaluated models. It also shows increasing accuracy as the game progresses, indicating its ability to integrate newly revealed information. For speech generation, WolfLLM significantly outperforms the same-size base model, while still suggesting room for further scaling. Ablation studies further show that identity inference and speech generation are mutually beneficial.

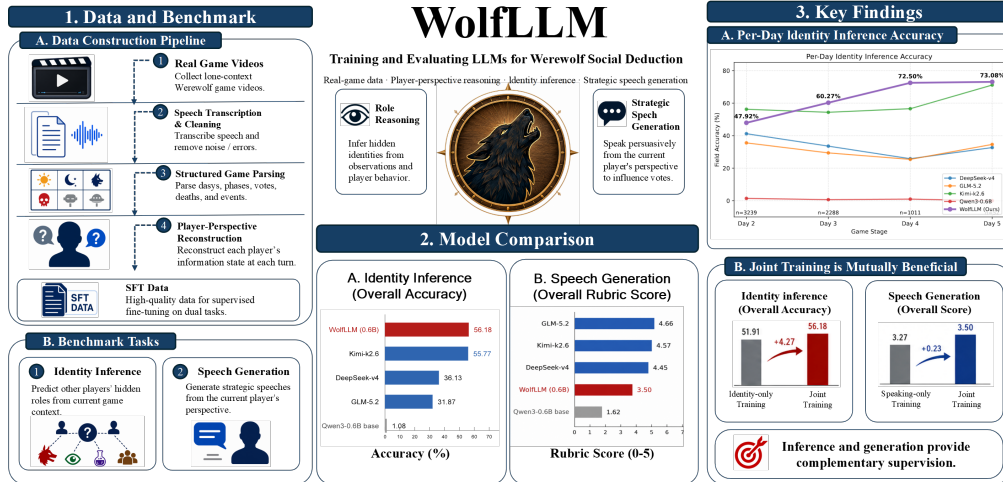


Figure 1: WolfLLM Teaser

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1 Introduction

Large Language Models (LLMs) have achieved remarkable progress across a wide range of natural language tasks, including mathematical reasoning, code generation, instruction following, and multi-turn dialogue. However, most existing benchmarks evaluate models in relatively static settings, where the task goal is explicit, the information is fully observable, and the correct answer is often fixed. Such evaluation protocols are insufficient for measuring model abilities in dynamic social environments, where agents must track long-term contexts, reason under incomplete information, and communicate strategically with other participants.

Social deduction games provide a natural testbed for evaluating these abilities. In particular, **Werewolf** is a representative incomplete-information game that requires players to infer hidden identities, interpret multi-turn speeches, analyze voting behaviors, and adapt to newly revealed game events. Unlike conventional reasoning tasks, Werewolf involves asymmetric information: different players have access to different private information depending on their roles. For example, wolves know their teammates, the Seer obtains inspection results, while villagers can only infer identities from public speeches and events. Therefore, a model that plays Werewolf must not only understand the public dialogue, but also reason and speak from a specific player perspective.

Despite the strong general capabilities of modern LLMs, Werewolf remains challenging for them. First, the game requires long-context tracking and dynamic belief updating. As the game progresses, new speeches, votes, deaths, role claims, and contradictions continuously change the available evidence. A capable model should be able to integrate such information over time and update its identity judgments accordingly. Second, Werewolf requires strategic language generation. A good speech is not merely fluent; it must be consistent with the speaker’s role, available information, faction objective, and current game state. Third, existing resources for this domain are limited. Synthetic self-play data often lacks the diversity, noise, and strategic complexity of real human games, while real-game data is difficult to convert into structured training and evaluation samples due to multi-speaker speech, noisy transcripts, complex game stages, and hidden-role information.

To address these challenges, we introduce a framework for constructing, training, and evaluating LLMs in the Werewolf domain. We collect real Werewolf game videos and convert unstructured multi-speaker interactions into structured game trajectories. During this process, we recover player numbers, game stages, board configurations, true roles, public events, and key night actions. More importantly, we reconstruct **player-perspective information states**, ensuring that each training or evaluation sample only contains the information legally available to the current player at that moment. This design preserves the incomplete-information nature of Werewolf and prevents unrealistic information leakage.

Based on the constructed data, we formulate two core tasks: *identity inference* and *speech generation*. Identity inference evaluates whether a model can predict other players’ roles from the current game context and player-perspective information. Speech generation evaluates whether a model can produce reasonable and strategic utterances under the same information constraints. We further train **WolfLLM**, a Werewolf-specialized model, using supervised fine-tuning on both tasks. Experiments show that WolfLLM substantially improves identity inference over its base model and achieves the best overall field-level accuracy among the evaluated models. In addition, WolfLLM exhibits a clear upward trend in identity inference accuracy as the game progresses, suggesting its ability to integrate newly available evidence. For speech generation, WolfLLM significantly outperforms the same-size base model, although it still lags behind much larger general-purpose LLMs, indicating a promising direction for further scaling.

The main contributions of this paper are summarized as follows:

- We construct a real-game Werewolf corpus by converting raw multi-speaker game videos into structured game trajectories with player numbers, stages, roles, events, and board configurations.
- We introduce player-perspective information reconstruction to preserve the incomplete-information structure of Werewolf and prevent information leakage during training and evaluation.
- We build a dedicated Werewolf evaluation benchmark covering two complementary tasks: identity inference for role reasoning and speech generation for strategic communication.

- We train WolfLLM with supervised fine-tuning and show that it achieves strong identity inference performance, improves over the base model in speech generation, and benefits from joint training of inference and generation tasks.

2 Related Work

2.1 Large Language Models and Complex Reasoning

Large language models have been widely studied for their reasoning capabilities. Existing benchmarks often evaluate models through mathematical problem solving, code generation, logical reasoning, and long-context question answering. These tasks test a model’s ability to perform symbolic reasoning, multi-step inference, and information retrieval from long contexts. However, they usually have well-defined inputs and outputs, and the correctness of answers can often be directly verified. As a result, they cannot fully capture the challenges of real-world interactive scenarios where agents must reason under uncertainty, model other agents, and communicate strategically.

2.2 Language Intelligence in Social Deduction Games

Social deduction games such as Werewolf, Avalon, Mafia, and Blood on the Clocktower require players to reason under hidden identities and incomplete information. Unlike conventional text reasoning tasks, these games emphasize long-term information integration, player-perspective modeling, deception detection, and strategic communication. Recent studies have begun to explore the use of LLM agents in social deduction games. However, most existing work focuses on prompting, agent simulation, or online game-playing performance, while large-scale supervised training data from real human games and standardized offline benchmarks remain underexplored.

2.3 Supervised Fine-Tuning for Domain-Specific LLMs

Supervised fine-tuning is an effective approach for adapting general-purpose LLMs to specific domains and tasks. By constructing high-quality instruction data, a model can learn domain-specific terminology, task formats, reasoning patterns, and response styles. In the Werewolf setting, supervised fine-tuning can help a model learn not only game-specific vocabulary and speaking conventions, but also the reasoning patterns required for hidden-role inference and strategic speech generation. Therefore, constructing a high-quality Werewolf SFT corpus is a crucial step toward improving the social reasoning ability of LLMs in this domain.

3 Method

3.1 Overview

Our method consists of three main components: real-game Werewolf data construction, evaluation task design, and supervised fine-tuning. First, we construct structured game data from real Werewolf game videos, converting unstructured multi-speaker speech interactions into game records that contain player numbers, game stages, role identities, public events, and player-perspective information. Second, based on the structured game records, we design evaluation tasks tailored to the Werewolf scenario, aiming to assess models’ abilities in strategic speech generation and hidden-role inference. Finally, we convert the reconstructed game states into supervised fine-tuning samples and train WolfLLM, a specialized model for Werewolf gameplay.

3.2 Data Construction

Werewolf is a typical incomplete-information game, where real matches involve complex natural language reasoning, identity deception, factional competition, and dynamic information updates. To obtain realistic and strategically meaningful training and evaluation data, we build our dataset from high-quality Werewolf game videos. The overall data construction pipeline consists of four steps.

Step 1: Raw Game Video Collection We first collect real Werewolf game videos as the raw data source. Each video usually contains a complete match, covering night actions, daytime speeches,

sheriff election, voting and exile, last words, and final role revelation. Compared with manually synthesized data, real games better reflect how players reason, persuade, deceive, and react under complex and evolving game states.

During data selection, we prioritize games with complete procedures, clear speech, standardized referee announcements, and high-quality player discussions. Complete games provide a continuous information flow from the beginning to the end of a match, allowing later samples to preserve the temporal evolution of the game state. Clear speeches and standardized referee announcements also help reduce errors in automatic transcription and structured parsing.

Step 2: Speech Transcription and Text Cleaning After collecting the raw videos, we use an automatic speech recognition system to transcribe the audio into text. Since Werewolf games usually involve multi-speaker turn-taking, informal expressions, repetitions, pauses, interruptions, and background noise, the raw ASR transcripts cannot be directly used for model training and evaluation. Therefore, we further perform text cleaning and normalization.

Specifically, we remove noise unrelated to the game, correct common recognition errors, merge consecutive utterances from the same player, and organize referee announcements and player speeches in chronological order. Werewolf games contain many domain-specific terms, such as “sheriff election,” “Seer,” “fake claim,” “golden water,” “silver water,” “wolf pit,” and “taking sides,” which are often misrecognized by general ASR systems. To address this issue, we incorporate a Werewolf-specific terminology list and common ASR error patterns during the cleaning process, so as to preserve the reasoning-relevant information in the original games as much as possible.

Step 3: Structured Game Parsing After text cleaning, we further parse the natural language transcripts into structured game records. The goal of this step is to recover complete game trajectories from unstructured transcripts, including player numbers, speaker identities, game stages, board configurations, true roles, voting results, death information, and key night actions.

First, we establish the mapping between speakers and player numbers based on referee announcements and speaking order. Second, we parse the board configuration of each game and determine the true role of each player by using final role revelation, night action results, and identity-related statements. Third, we segment the game into different stages according to referee announcements and game procedures, such as sheriff election, daytime discussion, voting and exile, last words, and night phases. Finally, we extract key events such as voting, deaths, inspections, saves, poisons, and exiles, and construct a temporally evolving game state.

Through this step, raw multi-speaker speech transcripts are converted into temporally ordered game trajectories with explicit event structures, providing the foundation for player-perspective modeling, training sample construction, and benchmark construction.

Step 4: Player-Perspective Information Reconstruction The core challenge of Werewolf lies in incomplete information: different players have access to different information at the same time. For example, wolves know their teammates, the Seer knows inspection results, the Witch knows potion usage, while ordinary villagers can only infer hidden identities from public speeches, votes, and death information. If global true identities are directly provided during training or evaluation, information leakage would occur, making the samples inconsistent with realistic gameplay conditions.

Therefore, we further reconstruct player-perspective information states. For each speaking turn, we only retain the information that the current player can legally access at that moment. Public information includes previous player speeches, referee announcements, voting results, death information, and exile results. Private information depends on the player’s role, such as wolf teammates and night-kill decisions for wolves, inspection results for the Seer, and potion usage for the Witch.

Player-perspective information reconstruction prevents the model from accessing future information or information unavailable to the current player, ensuring that both training and evaluation follow the information constraints of real Werewolf games. This mechanism also forces the model to perform identity inference and strategic communication under limited information, thereby more faithfully reflecting the decision-making environment faced by human players.

3.3 Benchmark and Evaluation Metrics

To systematically evaluate large language models in the Werewolf scenario, we construct a dedicated benchmark based on the structured game records. Unlike general language evaluation, Werewolf requires models not only to understand textual content, but also to perform identity inference, strategy selection, and faction-consistent language generation under dynamic and incomplete-information game states. Our benchmark consists of two complementary tasks: identity inference and speech generation.

Identity inference. Given the current game stage, historical speeches, public events, and player-perspective information, the model is required to infer the possible identities of other players. This task evaluates whether the model can integrate multi-turn speeches, voting behaviors, public game events, and role logic to make accurate identity judgments. Since the true roles are available in the structured game records, we report two automatic metrics. *Field Accuracy* measures whether each predicted identity field is correct, while *Exact Accuracy* measures whether the entire identity prediction for a sample exactly matches the ground truth.

Speech generation. Given the current player’s role, the game stage, historical public information, and player-perspective information, the model is required to generate a reasonable speech for the current turn. This task evaluates whether the model can understand the current game state and choose an appropriate communication strategy according to its role, faction objective, and available information. For example, villager-aligned players should analyze the situation and identify suspicious players, while wolf players need to hide their identities, mislead the good side, or protect their teammates.

Since speech generation is open-ended and cannot be reliably evaluated by exact string matching, we adopt a criterion-level rubric-based evaluation protocol. The rubric evaluates each generated speech from five dimensions: hidden-role reasoning, role and faction consistency, strategic value, factual faithfulness and hallucination control, and naturalness and persuasiveness.

Specifically, each dimension contains five fine-grained criteria. For each criterion, the evaluator assigns a judgment of *Yes* or *No*. A *Yes* judgment receives one point, while a *No* judgment receives zero points. If a criterion is not applicable to the current player’s identity or role, it is marked as *Not Applicable* and automatically receives one point. For example, the criterion about wolf-side deception is not applicable when the current player is not a wolf, and the criterion about power-role skill usage is not applicable when the current player is not a power role. Therefore, each dimension receives an integer score from 1 to 5.

The five rubric dimensions are defined as follows. **Hidden-role reasoning** evaluates whether the speech makes concrete suspicion, trust, or identity-related judgments, grounds such judgments in public history or private information, reasons about relations among multiple players, avoids logically incompatible identity claims, and updates beliefs when new information appears. **Role and faction consistency** evaluates whether the speech is consistent with the speaker’s true role and faction objective, helps the good side identify wolves or clarify information when the speaker is good, helps wolves conceal themselves or mislead the good side when the speaker is a wolf, properly handles power-role information such as inspections, potions, or hunter threat when applicable, and avoids unnecessary exposure of hidden identity or teammate information. **Strategic value** evaluates whether the speech has a clear communication goal, provides executable directions such as whom to vote, inspect, trust, or attack, benefits the speaker’s faction under the current situation, avoids generic or template-like content, and performs reasonable risk control. **Factual faithfulness and hallucination control** evaluates whether the speech avoids fabricating public events, treating uncertain inferences as confirmed facts, inventing unavailable private information, fabricating rules or role abilities, and creating non-existent player relations or behavioral evidence. **Naturalness and persuasiveness** evaluates whether the speech is fluent, Werewolf-like, internally consistent, concise, focused on the current game state, and persuasive to other players.

Let S_R , S_C , S_T , S_H , and S_N denote the scores of hidden-role reasoning, role and faction consistency, strategic value, factual faithfulness and hallucination control, and naturalness and persuasiveness, respectively. The overall speech generation score is computed as:

$$\text{Overall} = 0.25S_R + 0.25S_C + 0.20S_T + 0.20S_H + 0.10S_N. \quad (1)$$

Together, identity inference and speech generation provide a benchmark for evaluating both verifiable role reasoning and strategic communication in incomplete-information Werewolf games.

3.4 Supervised Fine-Tuning

Based on the structured game records and player-perspective information states, we further construct supervised fine-tuning data for training WolfLLM. Each training sample consists of an instruction, an input context, and a target output. The input context includes the current game stage, historical public information, the current player’s role, and the private information available to that player at the current moment. The target output corresponds to either the real player speech or the identity inference label, depending on the task type.

For the speech generation task, we use the real speeches produced by players in the corresponding turns as supervision signals. The model is required to learn how to generate natural language speeches that are consistent with the given game state, player identity, faction, and strategy. This task helps the model learn domain-specific expression patterns, reasoning styles, identity deception, and factional collaboration strategies in Werewolf games.

For the identity inference task, we construct supervision targets based on the structured role labels, requiring the model to predict the possible identities of other players under the current information condition. This task strengthens the model’s ability to model the relationships among historical speeches, voting behaviors, night events, and role logic. Compared with speech generation, identity inference provides more explicit supervision signals and helps improve the model’s judgment of hidden game states.

Finally, we decompose each complete game into multiple player-perspective samples, where each relevant speaking turn can produce one or more training instances. In this way, raw Werewolf videos are transformed into a large-scale instruction-tuning corpus. WolfLLM is then supervised fine-tuned on this corpus to learn game-state understanding, hidden-role inference, and strategic speech generation under incomplete information.

4 Experiments

4.1 Implementation Details

We use Qwen3-0.6B as the base model of WolfLLM and fine-tune it on the constructed Werewolf supervised fine-tuning corpus. The training corpus contains both speech generation samples and identity inference samples, formatted uniformly as instruction-following data. The model is optimized with the standard autoregressive language modeling objective.

We train WolfLLM for 2 epochs with a learning rate of 2×10^{-5} and a batch size of 1. Training is conducted on 8 NVIDIA A800 GPUs. Unless otherwise specified, the same input format is used for both training and evaluation, where each sample contains the current game stage, historical speeches, public events, role-related information, and the current player’s perspective.

For comparison, we evaluate several strong general-purpose LLMs, including Kimi-k2.6, GLM-5.2, and DeepSeek-v4-Flash. In addition, we include the original Qwen3-0.6B base model as a baseline to measure the effect of Werewolf-specific supervised fine-tuning. All models are evaluated on the same benchmark tasks and with the same input contexts to ensure a fair comparison.

4.2 Benchmark and Evaluation Metrics

We construct a dedicated Werewolf benchmark to evaluate social reasoning ability in two aspects: identity inference and speech generation.

For **identity inference**, the model is given the current game context and player-perspective information, and is required to predict the hidden identities of other players. We compare the predictions with ground-truth role labels and report two metrics. *Field Accuracy* measures whether each predicted identity field is correct, while *Exact Accuracy* measures whether the entire identity prediction for a sample exactly matches the ground truth.

For **speech generation**, the model is given the current game context, the speaking player’s identity, and the information available to that player. It is then required to generate the speech for the current turn. Since speech quality cannot be reliably evaluated by exact string matching, we adopt a rubric-based evaluation protocol. The rubric considers whether the generated speech understands the current game state, follows the player’s perspective, remains consistent with the player’s role, provides reasonable logic, avoids unavailable information, and demonstrates strategic communication.

4.3 Evaluation Results

We evaluate WolfLLM from three aspects: identity inference, reasoning across game stages, and speech generation quality. The results show that WolfLLM achieves strong advantages in hidden-role inference, exhibits a clear ability to integrate newly revealed game information, and substantially improves speech generation over the base model despite still lagging behind much larger general-purpose LLMs.

Model	Params	Day 2	Day 3	Day 4	Day 5	Overall
Samples	–	3239	2288	1011	52	6590
DeepSeek-v4	284B	41.22%	33.57%	25.82%	32.69%	36.13%
GLM-5.2	753B	35.57%	29.41%	25.42%	34.62%	31.87%
Kimi-k2.6	1.1T	56.25%	54.37%	56.58%	71.15%	55.77%
Qwen3-0.6B	0.6B	1.42%	0.66%	0.99%	0.00%	1.08%
WolfLLM (Ours)	0.6B	47.92%	60.27%	72.50%	73.08%	56.18%

Table 1: Per-day identity inference results from Day 2 onward. Samples denotes the number of field-level predictions, and all reported results are field-level accuracy. Params denotes the model parameter size.

Identity inference. As shown in Table 1, WolfLLM achieves the best overall field-level accuracy on the identity inference task. Specifically, WolfLLM obtains an overall accuracy of 56.18%, outperforming strong general-purpose LLMs such as Kimi-k2.6, DeepSeek-v4, and GLM-5.2. Compared with the original Qwen3-0.6B base model, whose overall accuracy is only 1.08%, WolfLLM brings a substantial improvement of 55.10 percentage points. This demonstrates that general-purpose language ability alone is insufficient for Werewolf identity reasoning, while domain-specific supervised fine-tuning can effectively inject game-specific knowledge, role logic, and player-perspective reasoning ability into a compact model.

Reasoning across game stages. We further analyze how model performance changes as the game progresses. In Werewolf, later stages provide more informative evidence, such as accumulated speeches, voting behaviors, night results, deaths, role claims, and contradictions among players. Therefore, a model with strong identity inference ability should be able to leverage newly available information and improve its judgments over time.

As shown in Figure 2, WolfLLM exhibits a clear upward trend from Day 2 to Day 5, improving from 47.92% to 73.08% in field-level accuracy. This indicates that WolfLLM can effectively integrate accumulated game evidence and continuously refine its identity inference as more information becomes available. In contrast, general-purpose LLMs do not show a similarly stable improvement across game stages. These results suggest that WolfLLM learns not only Werewolf-specific terminology and role logic, but also the ability to update identity judgments throughout the game.

Speech generation. Table 2 reports the rubric-based evaluation results for the Werewolf speech generation task. WolfLLM achieves an overall score of 3.50, substantially outperforming the Qwen3-0.6B base model, which only obtains 1.62. This improvement shows that Werewolf-specific supervised fine-tuning effectively enhances the model’s ability to generate speeches that are more consistent with the current role, game context, and strategic objective.

Meanwhile, WolfLLM still lags behind much larger general-purpose LLMs such as GLM-5.2, Kimi-k2.6, and DeepSeek-v4. This gap is expected, since WolfLLM is built on a compact 0.6B model, while high-quality Werewolf speech generation requires both domain-specific reasoning and strong

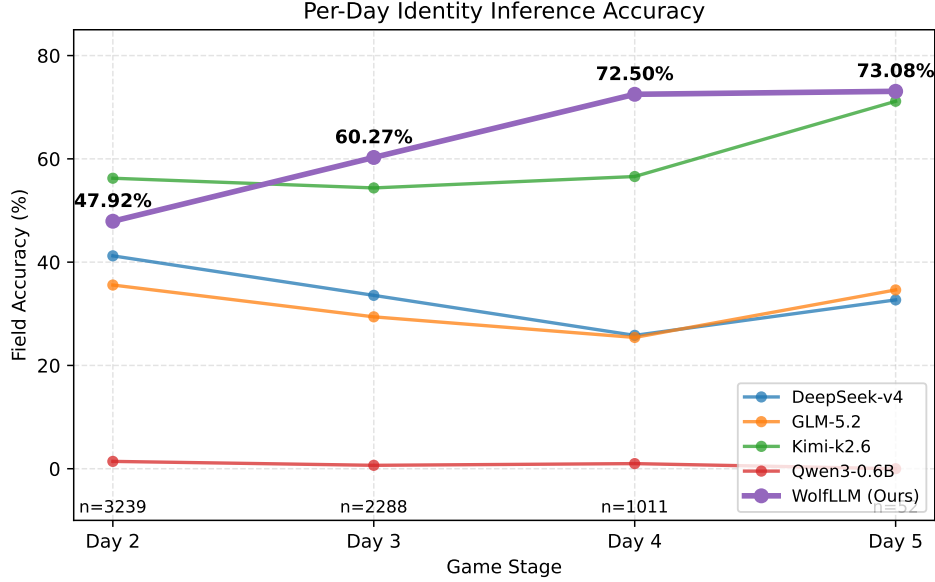


Figure 2: Per-day identity inference accuracy from Day 2 to Day 5. WolfLLM shows a consistent improvement as the game progresses, indicating its ability to integrate newly revealed information and update identity predictions over time.

Table 2: Rubric-based evaluation results on the Werewolf speech generation task. The five dimensions evaluate identity reasoning, faction consistency, strategic value, faithfulness, and naturalness, respectively.

Model	Identity Reason.	Faction Consist.	Strategy Value	Faithful. Control	Natural. Persuad.	Overall
GLM-5.2	4.84	4.61	4.57	4.59	4.68	4.66
Kimi-k2.6	4.85	4.54	4.62	4.18	4.64	4.57
DeepSeek-v4	4.70	4.55	4.29	4.11	4.57	4.45
Qwen3-0.6B	1.16	2.07	0.23	3.01	1.59	1.62
WolfLLM (Ours)	3.53	3.89	2.88	3.76	3.14	3.50

general language modeling ability. Therefore, the results suggest a clear scaling direction: with larger backbone models and more high-quality Werewolf training data, the proposed training framework has the potential to further improve strategic speech generation.

4.4 Ablation Study

Training Setting	Identity Reason.	Faction Consist.	Strategy Value	Faithful. Control	Natural. Persuad.	Overall
Speaking-only Training	3.77	3.71	2.71	3.13	2.29	3.27
Joint Training	3.53	3.89	2.88	3.76	3.14	3.50
Δ	-0.24	+0.18	+0.17	+0.63	+0.85	+0.23

Table 3: Ablation results on the Werewolf speech generation task. Joint Training uses both speech generation and identity inference data, while Speaking-only Training uses only speech generation data. Δ denotes the score difference between Joint Training and Speaking-only Training.

Multi-task synergy. To examine whether speech generation and identity inference can benefit from each other, we conduct an ablation study with two additional training settings: an identity-only model

Training Setting	Day 2	Day 3	Day 4	Day 5	Overall
Samples	3239	2288	1011	52	6590
Identity-only Training	46.03%	54.33%	63.80%	80.77%	51.91%
Joint Training	47.92%	60.27%	72.50%	73.08%	56.18%
Δ	+1.89	+5.94	+8.70	-7.69	+4.27

Table 4: Ablation results on the identity inference task. Samples denotes the number of field-level predictions, and field-level identity prediction accuracy is reported. Joint Training uses both speech generation and identity inference data, while Identity-only Training uses only identity inference data. Δ denotes the percentage-point difference between Joint Training and Identity-only Training.

trained only on identity inference data, and a speaking-only model trained only on speech generation data. The results are shown in Table 3 and Table 4.

For speech generation, joint training improves the overall score from 3.27 to 3.50. Although the speaking-only model performs slightly better on hidden role reasoning, joint training achieves higher scores on faction consistency, strategy value, fidelity and hallucination control, and naturalness and persuasiveness. This suggests that identity inference provides useful supervision for speech generation. In Werewolf, a strong player must first form accurate judgments about other players’ identities before producing logically coherent and strategically valuable speeches. Therefore, learning identity inference helps the model better understand the overall game situation, which in turn improves the quality of its generated speeches.

For identity inference, joint training also improves the overall field accuracy from 51.91% to 56.18%. The gains are particularly clear from Day 2 to Day 4, indicating that speech generation training provides additional benefits for identity prediction. We hypothesize that speech generation acts as a form of implicit reasoning supervision: to generate a reasonable speech, the model needs to learn how players express suspicion, defend themselves, build logical chains, and respond to game-state changes. These patterns implicitly teach the model why a player may correspond to a certain identity, thereby improving its identity inference ability.

Overall, the ablation results show that speech generation and identity inference are mutually beneficial in Werewolf modeling. Identity inference strengthens the model’s understanding of the game state and improves its strategic expression, while speech generation provides implicit reasoning signals that help the model make more accurate identity judgments.

5 Conclusion

In this paper, we present WolfLLM, a specialized language model for Werewolf, an incomplete-information social reasoning game. We construct structured game records from real Werewolf videos, reconstruct player-perspective information states, and build supervised data for both identity inference and speech generation. Experiments show that WolfLLM substantially improves identity inference over the base model and achieves competitive performance against much larger general-purpose LLMs, especially in later game stages where more evidence becomes available. Moreover, ablation studies demonstrate that identity inference and speech generation are mutually beneficial, suggesting that strategic language generation and game-state reasoning should be learned jointly. Although the current 0.6B model still lags behind large general-purpose LLMs in speech generation, its clear improvement over the base model indicates the effectiveness of our data construction and training framework, as well as the potential of further scaling model capacity and high-quality Werewolf data.